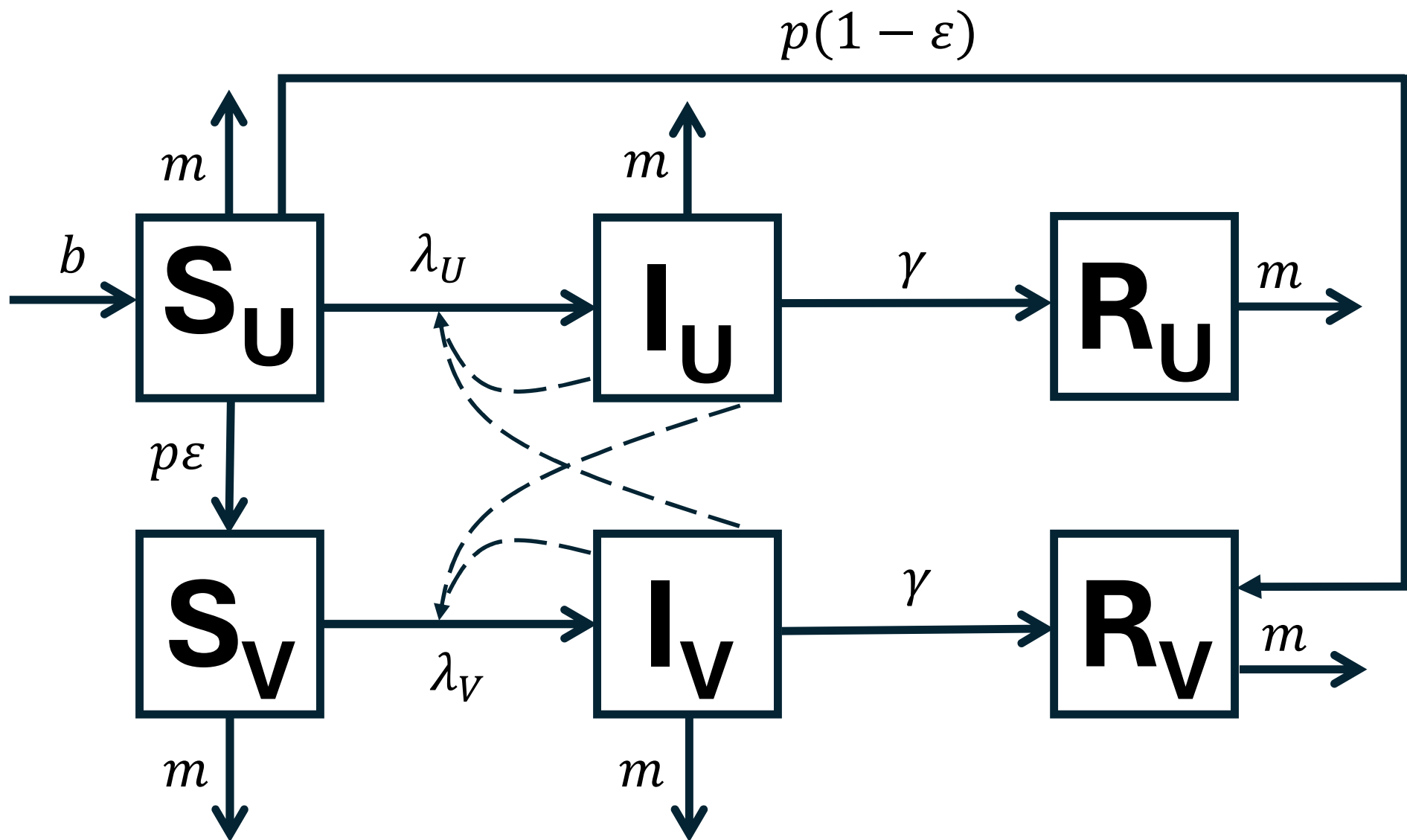




$$\lambda_U = \beta \left(\phi \frac{I_U}{N_U} + (1 - \phi) \frac{I_U + I_V}{N_U + N_V} \right)$$

$$\lambda_V = \beta \left(\phi \frac{I_V}{N_V} + (1 - \phi) \frac{I_U + I_V}{N_U + N_V} \right)$$